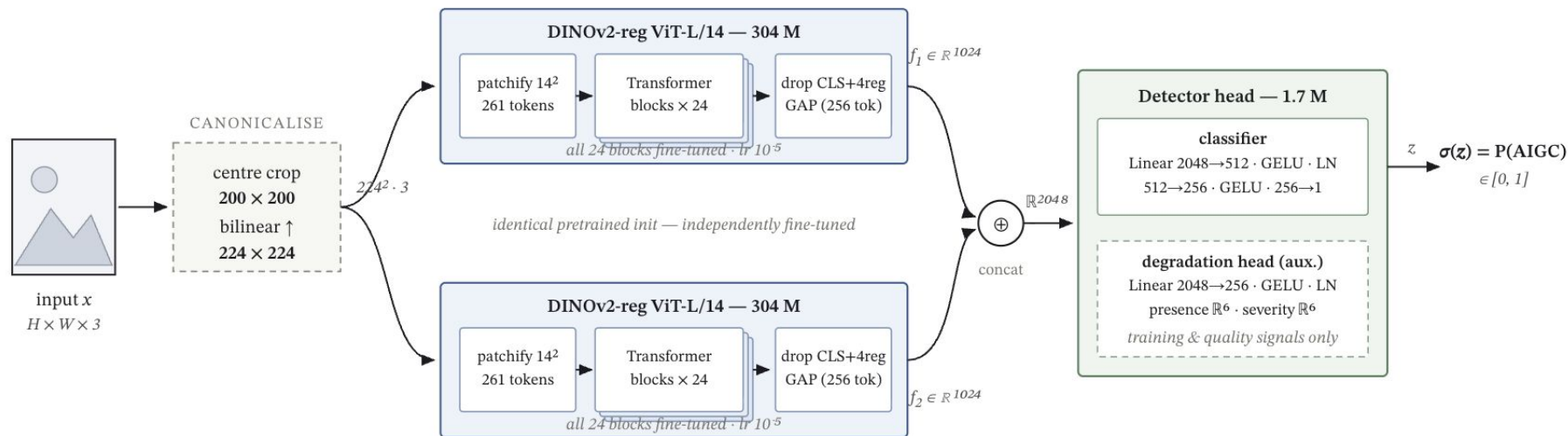


The Dual-224 Detector — Inference

Deterministic canonicalisation, two independently fine-tuned towers, one 2048-d readout.

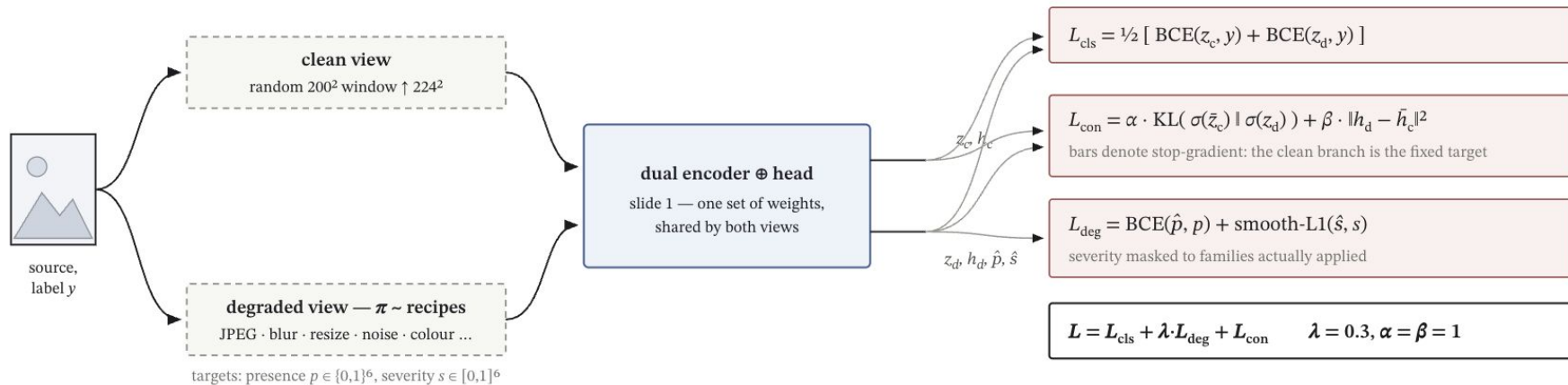


610.5M parameters = $2 \times 304.4\text{M towers} + 1.7\text{M head} \cdot \text{bfloat16 inference} \cdot \text{deterministic centre window}$, so the same file always scores identically

demo AUC 0.9989 $\cdot$ TPR@1%FPR 0.976 **transfer** 0.9963 $\cdot$ 0.983

The Dual-224 Detector — Training

Clean and degraded views of every source, one shared model, three losses.



head lr 10^{-3} · towers lr 10^{-5} (discriminative) · cosine + 3% warmup · bf16 autocast · gradient-checkpointed · 350,663 images · 64 sources/batch, clean $\leftrightarrow$ degraded pairs

The towers start from identical pretrained weights and diverge through the asymmetric gradients of their head halves. The degradation head is supervision, not signal — it never touches the score.

one epoch $\approx$ 1 h 50 m on 4×RTX4090 · dual_plan_d24/checkpoint_ep1.pt

Table1. Detection performance of the dual-224 detector across evaluation splits. All numbers are measured after a single epoch of fine-tuning (checkpoint `dual_plan_d24/checkpoint_ep1.pt`), scoring from pixels under the deterministic centre-window canonicalisation (200×200 crop $\uparrow 224^2$), `bfloat16` inference. $\text{TPR}@1\%\text{FPR}$ is the true-positive rate at the threshold that passes 99% of the split’s authentic images. Generator counts are fake-image counts; the real pools are described in the notes.

Split	AUC	TPR@1%FPR	Dataset size (Real / Fake)	Model families & counts
val^a <i>internal validation</i>	0.9654	0.7348	27,822 / 27,846	Klein-4B t2i (5,200) · NTIRE val (4,819) · MAE (3,500) · MAGE (3,500) · VQVAE (3,500) · VQGAN (3,500) · Klein-4B ref-image (2,600) · NTIRE val-hard (1,227)
test_transfer^b <i>unseen generators</i>	0.9963	0.9825	16,098 / 16,098	Würstchen t2i (7,808) · Sana-1600M t2i (7,800) · Google NanoBanana 2 Lite (100) · Ideogram 4.0 Turbo (100) · ByteDance Seedream 4.5 (98) · Meta Muse Image (97) · OpenAI GPT-Image-2 (95)
demo^c <i>fully held-out</i>	0.9989	0.9758	4,998 / 8,843	DALL·E 3 (8,843)

^a Real pool: COCO train2017, Open Images (incl. V7), NTIRE authentic; carries the corpus’s distorted/hard variants (NTIRE val-hard is the hardest mass, AUC 0.8776 on its slice), which is why val reads well below the two evaluation splits.

^b Real pool: Open Images V7 (31,216 of the pairs) plus small COCO/NTIRE/WildFake slivers. Every generator in this split is excluded from training; the five commercial APIs (490 images) are the hardest transfer slice.

^c Real pool: COCO val2017. Both sides fully held out from training and validation; accuracy at the 0.5 threshold is 99.3% on real and 96.6% on DALL·E 3.

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Table 2. Robustness of the dual-224 detector under the degradation grid, against the single-tower baseline. AUC on 2,490 test_transfer rows (stratified per source $\times$ label) under each post-canonicalisation degradation; every cell scores the same rows. The measurement is Δ against each model’s own clean reference — absolute AUC on this stratified subset is deliberately not comparable with the pooled full-split AUC (0.9963). The dual-224 model degrades less than the baseline (mean Δ -0.0031 vs. -0.0056 over the 16 template transforms) and starts higher on every single row.

Augmentation	Baseline AUC ^a	Dual-224 AUC	Δ vs clean (dual) ^b
clean <i>reference</i>	0.8891	0.9551	—
JPEG $q=90$	0.8947	0.9645	+0.0094
JPEG $q=70$	0.8908	0.9645	+0.0094
JPEG $q=50$	0.8865	0.9550	-0.0001
JPEG $q=30$ <i>brief’s grid</i>	0.8764	0.9414	-0.0137
JPEG $q=10$	0.8478	0.8949	-0.0601
Blur $\sigma=0.5$	0.8831	0.9673	+0.0122
Blur $\sigma=1.0$ <i>held-out severity</i>	0.8719	0.9585	+0.0035
Blur $\sigma=2.0$	0.8810	0.9500	-0.0051
Resize 0.5 $\times$	0.8910	0.9600	+0.0049
Resize 0.25 $\times$	0.8858	0.9353	-0.0198
Noise $\sigma=0.02$	0.8977	0.9584	+0.0033
Noise $\sigma=0.05$	0.8880	0.9565	+0.0015
Noise $\sigma=0.10$	0.8732	0.9411	-0.0140
Jitter brightness $\pm 20\%$	0.8809	0.9488	-0.0062
Jitter contrast $\pm 20\%$	0.8842	0.9617	+0.0066
Jitter saturation $\pm 20\%$	0.8918	0.9572	+0.0021
Jitter all three $\pm 20\%$ <i>brief’s grid</i>	0.8766	0.9557	+0.0006
Centre crop 80%	0.8869	0.9595	+0.0045
Mean 16 template transforms^c	0.8835	0.9521	-0.0031

^aBaseline: single DINOv2-reg ViT-L/14, crop200 $\uparrow$ 512 (outputs/unfreeze/d24/checkpoint.pt). Its own mean Δ vs clean over the same 16 transforms is -0.0056 .

^bAUC standard error $\approx \pm 0.005$ at $n=2,490$; $|\Delta| < 0.01$ (grey) is a tie; only coloured entries are outside noise. Green = better than clean, red = worse.

^cExcludes JPEG $q=30$ and combined jitter (the brief’s own grid cells) so the mean covers the sixteen template transforms.

Six ablation studies behind the shipped dual-224 detector, in the order the programme ran them. Tables 3–4 and 7–8 use the pre-registered selection metric **robust TPR@1%FPR**: the mean true-positive rate over the 19 degraded evaluation conditions at the threshold passing 99% of authentic images, measured on *val_internal* authentic vs. held-out-generator images. WildFake columns hold out two generator families from that corpus; AI-OV7 is a full transfer corpus whose generators are never trained on. Absolute values are comparable only within a column.

Table 3. Unfreezing the backbone is the largest single effect in the programme. One variable — how many of the 24 transformer blocks train — on DINOv2-reg ViT-L/14 under the crop policy, 20k probe corpus, 5 epochs, head lr 10^{-3} / tower lr 10^{-5} . The ladder rises monotonically with *no cliff at full depth* on both corpora, which is why the shipped model fine-tunes all 24 blocks.

Depth	Blocks	Trainable params	WildFake held-out <i>robust TPR@1%FPR</i>	Δ vs D0	AI-OV7 transfer <i>robust TPR@1%FPR</i>	Δ vs D0
D0 <i>frozen control</i> ^a	0	0	0.8023	—	0.3773	—
D1	1	12.6 M	0.8565	+0.0542	—	—
D2	2	25.2 M	0.8885	+0.0861	—	—
D4	4	50.4 M	0.9388	+0.1365	0.6069	+0.2297
D6	6	75.6 M	0.9554	+0.1531	0.6583	+0.2810
D8	8	100.8 M	0.9698	+0.1675	0.7054	+0.3281
D24 <i>shipped depth</i>	24	302.4 M	0.9841	+0.1818	0.7580	+0.3807

^a D0 is retrained with the tower frozen rather than copied from the cached-feature rung a3: this path runs the tower live in float32 where a3's head was fitted on float16 cached features, and on the same bank a3 scores 0.8300 vs. D0's 0.8023. Without the control that 0.0277 dtype artefact would be silently added to D1's depth effect.

Frozen features train a 0.92 M head — 0.3% of the tower. Every depth re-extracts its own evaluation bank (a tower whose weights moved no longer produces the cached features), and all depths share the manifest, conditions, policy and rows. D1/D2 were run on the WildFake arm only.

Table 4. Crop beats band on both fine-tuned towers, and the margin explodes on transfer. Two canonicalisation policies — **crop** (native-resolution 200² window, upsampled) vs. **band** (whole frame resampled to a nominal side, jitter 0.1) — each trained to full 24-block unfreeze on two different backbones. Same sign on two towers and two corpora makes this a property of the *policy*, not of either tower.

Tower	Policy	WildFake held-out	AI-OV7 transfer	Δ crop–band <i>WF / OV7</i>
DINOv2-reg ViT-L/14	crop	0.9841	0.7580	+0.0136 / +0.1251
	band	0.9705	0.6329	
SigLIP2-L/512	crop	0.8706	0.5477	+0.0123 / +0.0521
	band	0.8583	0.4956	

Band also fails the confound gate: it resamples away the generator’s native pixel grid and substitutes the resampler’s own, and a content-blind probe classifies band-canonicalised images at pooled AUC 0.6105 (0.9976 on SID_Set) where crop sits near chance (0.5081 / 0.6316). Band would destroy the real signal and supply a fake one.

Training loss mis-ranked the arms in *both* towers — SigLIP band reached a lower total loss than SigLIP crop (0.1110 vs. 0.1281) and scored worse held-out metrics, never on loss.

Table 5. The final candidates, and why the dual-224 ships solo. The upstream funnel — frozen probe over six backbones (DINOv2-reg wins), the depth ladder (Table 3: unfreeze everything), the policy ablation (Table 4: crop), and a 15-combination fusion lattice whose every ensemble *lost on transfer* — leaves two plan-scale arms and their ensembles. The dual-tower model wins or ties every cell; averaging in the weaker arm only dilutes it.

Model	val <i>AUC / TPR@1%</i>	test_transfer <i>AUC / TPR@1%</i>	demo <i>AUC / TPR@1%</i>
single tower, crop 200↑512 ^a	0.9177 / 0.6383	0.9897 / 0.9668	0.9956 / 0.8739
dual-224 <i>shipped</i>	0.9654 / 0.7348	0.9963 / 0.9825	0.9989 / 0.9758
ensemble, equal weights	0.9655 / 0.7159	0.9955 / 0.9810	0.9988 / 0.9764
ensemble, fitted $w = 0.9^b$	fit split	0.9961 / 0.9823	0.9989 / 0.9759

^aThe Table 3/4 winner (DINOv2-reg, all 24 blocks, crop 200) retrained at plan scale, nominal side 512 (outputs/unfreeze/d24/checkpoint.pt).

^bWeight on the dual model, swept on val only, then read once on the two evaluation splits. Even the fitted ensemble never beats the dual model alone — the earlier lattice finding (ensembles gain in-domain, lose on transfer, all 14 non-trivial combinations) repeats at plan scale.

Table 6. Eight-view test-time augmentation buys nothing worth 8× the inference cost. The shipped model scored clean vs. the mean of eight per-view sigmoid probabilities (identity, flips, mild JPEG/blur/resize views) over the full evaluation splits. Every delta is at or below the third decimal place, and on demo TTA is marginally *negative*. The submission scores single-view.

Split	Metric	Clean	8-view TTA	Δ
test_transfer <i>n</i> = 32,196	AUC	0.9963	0.9974	+0.0011
	TPR@1%FPR	0.9825	0.9825	+0.0001
	acc@0.5	0.9871	0.9875	+0.0003
demo <i>n</i> = 13,841	AUC	0.9989	0.9989	± 0.0000
	TPR@1%FPR	0.9758	0.9748	−0.0010
	acc@0.5	0.9760	0.9759	−0.0001

The one slice TTA moves is the hardest: the 490 commercial-API fakes in test_transfer go from 26.7% to 31.4% caught at threshold 0.5 — real, small, and the source of the +0.0011 AUC. At probe scale the properly temperature-refitted a6 rung told the same story (+0.0023 over a3, with the TTA temperature 1.9653 vs. single-view 1.9759). Entropy-minimising test-time *adaptation* (MEMO/TENT-style) was piloted separately and was neutral-to-harmful at every learning rate.

Table 7. The reconstruction (VAE) branch: helps frozen features, and the VQ autoencoder helps more. Each rung adds a 12-d round-trip descriptor (per-band reconstruction error, AEROBLADE’s LPIPS slot, FIRE’s mid-frequency bands) from one autoencoder to the frozen DINOv2-reg crop features, differing from its base by exactly one flag. One held-out family is VQGAN — and a continuous KL VAE has no structural reason to round-trip a VQ decoder’s output anomalously, which is exactly how the measurement comes out.

Rung	Reconstruction branch	Dim added	robust TPR@1%FPR	Δ vs a3
a3 <i>base</i>	none	—	0.7858	—
a4	SD 1.5 KL VAE	+12	0.8229	+0.0371
a4vq	VQ autoencoder	+12	0.8401	+0.0543
a4both	both	+24	0.8422	+0.0564

Measured on the frozen 20k probe bank (`docs/selection_probe_crop_dinov2regl_*.json`); a re-measurement on the depth ladder’s float32 bank compressed the margins to +0.0055 (a4both) and -0.0044 (a4), so the gain is bank-sensitive. Set beside Table 3’s +0.18 from unfreezing, the branch did not earn its two autoencoders in the shipped model, which carries no reconstruction features.

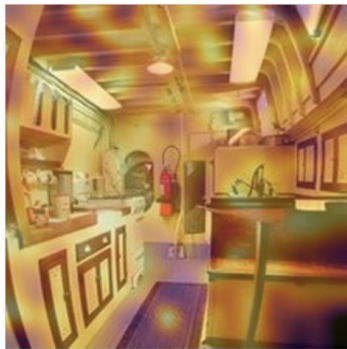
Table8. The frequency branch is a family property — and its control probe is the important row. aF adds a 4-d upsampling-periodicity descriptor (two NPR magnitudes plus horizontal/vertical contrast) to the frozen a3 features, under crop only: band resampling destroys the native pixel grid the descriptor reads. It helps the DINO lineage, does nothing for SigLIP, and — the control — carries almost no signal on its own, so the gain is complementary to the backbone rather than a low-level shortcut.

Arm <i>crop policy, 20k probe</i>	a3	aF = a3 + freq	Δ
DINOv2-reg ViT-L/14	0.7874	0.8317	+0.0443
DINOv2 ViT-L/14	0.7047	0.7355	+0.0308
SigLIP-SO400M	0.6399	0.6358	−0.0041
frequency-only head <i>control, no backbone</i>	—	0.0201 (<i>AUC 0.593</i>)	≈ chance

The control also ran the corpus’s sharpness/JPEG/noise proxies (3-d, TPR 0.0157) and both together (7-d, 0.0199) — all near chance, so aF’s +0.04 cannot be a relearned quality confound. In the fusion lattice the block behaved as a cheap substitute for a third tower (+0.0141 fused); as a standalone arm it loses, and like the reconstruction branch it is subsumed once the towers themselves fine-tune (Table 3), so the shipped model does not carry it.

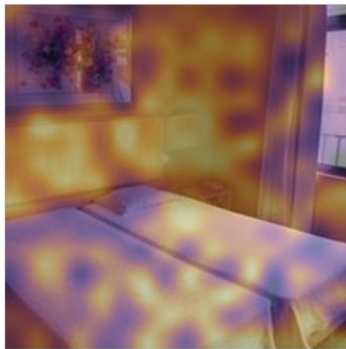
Figure 2. Where the evidence sits — per-patch maps of the shipped model’s worst errors. The detector head consumes the global average of the two towers’ 256 patch tokens; applying that same head to each token pair individually yields a 16×16 evidence map with no extra machinery — the map and the score are the same computation, one pooled and one not. Each tile shows the map (inferno, $\alpha=0.45$) blended over the canonicalised 224^2 view the score was actually computed on; the pooled probability under each tile reproduces the reported evaluation score to the third decimal. Colours are normalised *within* each image and read as “evidence concentrates here”, never as per-region probabilities.

FALSE POSITIVES *authentic photographs scored as generated — the three worst of 4,998 COCO reals*



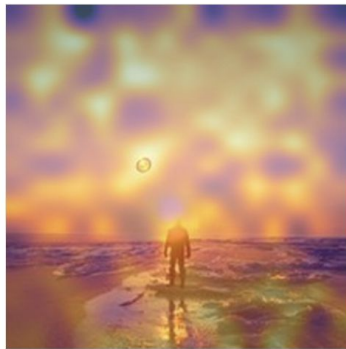
COCO val2017 · authentic

P(AIGC)=0.994



COCO val2017 · authentic

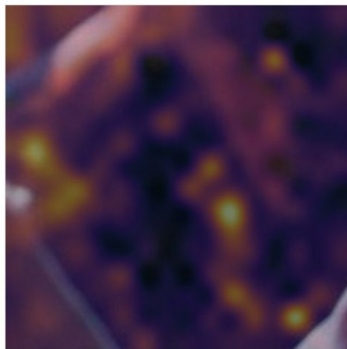
P(AIGC)=0.985



COCO val2017 · authentic

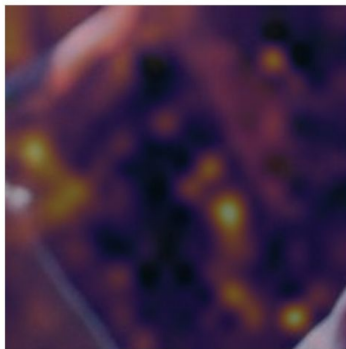
P(AIGC)=0.985

FALSE NEGATIVES, SEEN GENERATOR *DALL·E 3 images scored as real — the three worst of 8,843*



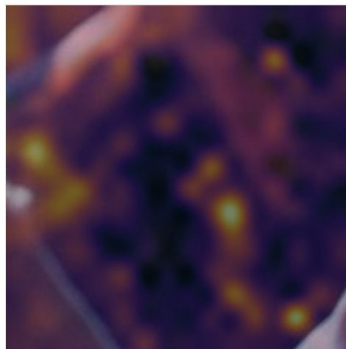
DALL·E 3

P(AIGC)=0.005



DALL·E 3

P(AIGC)=0.005



DALL·E 3

P(AIGC)=0.005

What the maps say about the failure modes

False positives concentrate on smooth, processed regions. REAL → FAKE The three worst reals are a heavily distorted wide-angle interior, a soft low-light bedroom, and a backlit sunset whose sky is pure glow — and in each the heat sits on the smooth ceiling panels, the denoised walls, the featureless bright sky. That is the computational-photography signature: denoise-then-sharpen pipelines, HDR fusion and lens correction leave smoothed noise floors and hallucinated micro-detail that sit statistically close to diffusion output.

Seen-generator false negatives are degradation cases. FAKE → REAL The worst DALL·E 3 misses are dark, heavily defocused frames. Compare the control row: on a typical caught DALL·E3 image the evidence is dense and *everywhere* — texture-level artefacts across the whole frame. Where blur and darkness destroy that high-frequency signal, the map goes flat: the evidence is physically absent, in both directions.

Unseen decoders fail flat, not mislocated. FAKE → REAL On the commercial-API misses (Ideogram 4.0 Turbo, NanoBanana 2 Lite, GPT-Image-2) the map is not pointing at the wrong place — it finds little evidence anywhere. The artefact distribution these decoders leave was never in training; this is the generalisation ceiling of an artefact-based detector, not a localisation bug. Newer open-weight candidates (Qwen-Image variants, CogView4) were outside our hardware and generation-time budget, so the slice stays a measured, honest weakness: 26.7% caught at threshold 0.5, though ranking (AUC) on the slice remains far better than the thresholded accuracy suggests.

Dataset conflation is handled upstream, imperfectly. BOTH Source-specific artefacts (resolution, format, capture pipeline) correlate with the label in every public corpus, so a model can learn provenance instead of generation. The deterministic centre-crop canonicalisation exists to break exactly that shortcut — the maps above are computed on those canonicalised pixels — but residual bias remains and the false-positive modes show where.